

RATUL HASAN

Robotics Enthusiast

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PROFESSIONAL SUMMERY

Innovative technologist specializing in robotics, autonomous drones, IoT and AI-driven systems. Experienced in developing scalable mobile applications and integrated hardware–software solutions across automation and digital education. Built autonomous rovers, bionic hand simulations, real-time LMS platforms and interactive games. Skilled in Flutter, full-stack development, embedded systems and intelligent system design.

KEY SKILLS

- Robotics
- Full-stack app development
- Wireframe Creation
- Game Development
- cloud computing
- 3D modeling
- Web development
- Graphics design
- Augmented reality
- simulation
- Cyber security
- Bash Scripting
- Linux echo-system
- server administration
- Ros (robot operating system)

PROFESSIONAL EXPERIENCE

2018 - Present **ONNESOK | FOUNDER**

Founded and led Onnesok, a technology platform focused on building scalable software solutions and immersive gaming experiences. Onnesok develops a portfolio of digital products, including innovative software tools and interactive mobile games, aimed at delivering engaging and high-performance digital experiences. Responsible for product vision, software architecture, project execution, and user experience design to bring modern digital applications from concept to launch.

2024 - Present **BRAC UNIVERSITY RS APP DEV INSTRUCTOR**

App Development Instructor at the Residential Campus of BRAC University, delivering intensive, hands-on training in mobile app development. Mentoring students through the full software development lifecycle—from system design and implementation to testing and deployment—while emphasizing clean architecture, teamwork, and industry-ready coding practices.

2025 - present **BRAC SKILL DEVELOPMENT PROGRAM | STUDENT DEVELOPER**

Currently contributing to the SDP ecosystem, developing and improving technical infrastructure and student engagement tools. Enhancing digital platforms to support learning, collaboration, and project management. Focused on creating efficient, user-friendly solutions that streamline workflows and improve overall student experience for job searching.

2025-present	LEARNERS BOOSTERS CO-FOUNDER & LECTURER Currently teaching Flutter and full-stack development, delivering hands-on lessons and projects to help students build scalable mobile and web applications. Design interactive learning materials, mentor students in coding best practices, and guide them in applying technical skills to real-world projects. Focused on fostering practical development expertise and problem-solving abilities.
2022 - 2025	ROBOWAY LABS FORMER CHIEF TECHNOLOGY OFFICER Former Chief Technology Officer (CTO) at Roboway Labs – Led the development of robotics, IoT, and AI-driven solutions, overseeing projects from concept to deployment. Directed cross-functional teams to deliver innovative products such as Pixi humanoid robot version 1 and version 2, Caarmate app, pet fiddler design and many more.
2024 - 2026	BIOSE RESEARCH TEAM LEAD Research team lead of project 1006 - Developed a functional bionic arm using Arduino, sensors, and servo motors, with real-time 3D simulation in VPython to replicate natural finger movements. Focused on modularity, precision, and educational applications.
2024 - 2025	ECOLLAB INTERNATIONAL FORMER FACILITATOR Facilitated youth-led climate initiatives across Bangladesh, Nigeria, Jordan, Indonesia, and Brazil, empowering young individuals through education, community engagement, and advocacy to drive meaningful climate action.
2022 - 2026	SECRETARY AT ROBU Served as Secretary of RPM, responsible for coordinating research and project activities, managing documentation, organizing meetings and facilitating communication to ensure smooth project execution and team collaboration.
2022 - 2023	BRACU DUBURI FORMER 3D AND MECHANICAL DESIGNER Designed and developed mechanical components for Bangladesh's first international AUV team, including torpedo and dropper systems, creating precise and reliable mechanisms for autonomous underwater missions.

EDUCATION

- **BACHELOR OF SCIENCE | COMPUTER SCIENCE AND ENGINEERING**
BRAC University (Currently studying)
- **HIGHER SECONDARY CERTIFICATE | SCIENCE**
Govt. Bangabandhu college
- **SECONDARY SCHOOL CERTIFICATE | SCIENCE**
S.M. Model Govt. high school

PROJECTS

- **BIONIC ARM**

Developed a functional bionic arm capable of simulating natural finger movements. Built the mechanical and electronic systems, and implemented VPython for real-time 3D visualization to enable accurate testing and feedback. Focused on modularity, responsiveness and usability to create an educational and practical prosthetic solution.

- **AXELIA**

Developed Axelia, a four-legged autonomous robot designed for stability, mobility, and versatile terrain navigation. Engineered its mechanical structure and control system to enable smooth walking, turning, and obstacle negotiation. Integrated real-time feedback mechanisms and programming to achieve precise motion control and adaptive movement. Focused on modular design and scalability for research, educational, and practical robotic applications.

- **PIXI V1 AND PIXI V2**

Developed Pixi, an interactive human sized robotic platform designed to engage users through responsive behaviors and real-time control. Engineered its hardware and software systems to enable smooth interaction, sensor-based feedback, and adaptive responses. Focused on modularity, usability, and educational applications to facilitate learning in robotics and programming.

- **AI DUINO**

Developed Aduino, a mobile application that transforms a smartphone into the control and sensor hub for a robotic rover. Leveraged the phone's internal sensors, including accelerometer, gyroscope, and camera, to enable real-time navigation, motion control, and environmental sensing. Focused on modularity and cost-effectiveness, reducing the need for additional hardware while providing a versatile platform for robotics experimentation and prototyping.

- **ENVIRONMENT CLEANING ROBOT**

Developed an autonomous environment cleaning robot designed to navigate spaces and perform cleaning tasks efficiently. Engineered its mechanical design, control systems, and software for obstacle detection, path planning, and task automation. Focused on reliability, adaptability, and scalability to create a practical solution for automated sanitation in various environments.

- **SHOOTERX**

ShooterX is an adrenaline-fueled, endless arcade shooting game. Your mission is to navigate the treacherous depths of space, dodging and destroying waves of asteroids and hostile enemy spaceships to ensure the survival of your kind. The game features beautifully rendered dynamic backgrounds, silky-smooth controls, and a difficulty curve that constantly challenges you. It's not just a game; it's an epic struggle for survival with endless fun!

- **ADVENTURE OF SUI**

Adventure of Sui is a captivating pixel-art RPG adventure game that takes you on a journey through vibrant forests, treacherous dungeons, and mysterious lands. Guide Sui as you solve puzzles, defeat enemies, and uncover hidden secrets in a beautifully crafted world. Featuring smooth controls, charming animations, and a difficulty curve that keeps every level engaging, the game combines exploration, strategy, and adventure for endless fun. It's not just a game; it's a journey filled with challenges, discovery, and excitement!

- **TEXLAB | ONLINE LATEX EDITOR**

TexLab is an online LaTeX editor designed to streamline the creation of professional-quality documents, research papers, and academic reports. It provides a user-friendly interface for writing, compiling, and previewing LaTeX documents in real-time, making it accessible for both beginners and advanced users. With features like syntax highlighting, template support, collaborative editing, and cloud-based project management, TexLab simplifies the process of producing well-formatted technical and academic content efficiently.

- **FARMBOT**

Developed FarmBot, an autonomous robotic system for precision farming. Engineered its hardware and software to perform automated planting, watering, and monitoring of crops, optimizing resource usage and crop yield. Focused on scalability, efficiency, and real-time data collection to support smart agriculture practices.

- **ROBU APP**

Developed ROBU App, a comprehensive mobile platform for the Robotics Club of BRAC University. Integrated features include resources, events, announcements, interactive tutorials, Arduino project integration, podcasts, and dynamic UI modules. Designed to enhance member engagement, facilitate learning, and support hands-on robotics projects. Focused on usability, modularity, and real-time content updates to create an all-in-one hub for the club community.

- **CAARMATE APP**

Developed CaarMate, a mobile app designed to facilitate carpooling and shared transportation. Implemented real-time ride matching, route optimization, and secure user authentication to connect drivers and passengers efficiently. Focused on enhancing accessibility, reducing transportation costs, and promoting eco-friendly commuting solutions.

- **CHICKEN COOP**

Developed Chicken Coop, a smart tracking system for monitoring empty rooms at BRAC University. Implemented real-time occupancy detection and status updates to help students and staff identify available rooms efficiently. Focused on usability, reliability, and integration with campus systems to optimize space management.

- **CO2 REACTOR**

Designed a CO₂ reactor system aimed at capturing and utilizing carbon dioxide efficiently. Engineered the system for optimal gas flow, reaction efficiency, and scalability. Focused on sustainable and practical solutions for carbon capture and environmental impact reduction.

- **BUX ANDROID APP**

Developed BUX, the mobile app for BRAC University's Learning Management System (LMS), enabling students and faculty to access courses, submit assignments, view grades, and receive announcements on-the-go. Focused on real-time updates, user-friendly interface, and seamless mobile accessibility to enhance the academic experience.

- **ARDUINO SERIAL COMMUNICATOR**

Developed an Arduino-based serial communication system to enable real-time data transfer between devices. Implemented efficient data encoding and decoding, ensuring reliable and low-latency communication for hardware projects. Focused on modularity, usability, and seamless integration with various sensors and actuators.

- **THEO JANSEN MECHANISM BASED EIGHT LEGGED ROBOT**

Developed an eight-legged walking robot inspired by the Theo Jansen mechanism, designed for smooth and stable locomotion. Engineered the mechanical structure and linkages to replicate natural walking patterns, optimizing movement efficiency and terrain adaptability. Focused on modularity, precision, and practical applications in robotics research and education.

- **OVIJAN ROBOT**

Developed Ovijan, an autonomous rescue robot designed to navigate disaster zones and assist in search-and-rescue operations. Engineered its mechanical structure, control systems, and sensor integration to detect obstacles, traverse challenging terrain, and locate victims. Focused on reliability, real-time responsiveness, and practical deployment for emergency response scenarios.

- **BONDHU BAJAR APP**

Developed Bondhu Bazar, a mobile application for a startup aimed at connecting buyers and sellers in a local marketplace. Implemented features for product listing, order management, secure transactions, and real-time notifications. Focused on user-friendly design, scalability, and seamless mobile experience to facilitate efficient local commerce.

- **PHOENIX DRONE**

Developed an autonomous mapping drone for precision aerial surveying and data collection. Integrated flight control, GPS navigation, and imaging systems to generate accurate maps and 3D models. Focused on real-time data processing, high accuracy, and applications in agriculture, infrastructure monitoring, and environmental analysis.

- **BLINKY**

Blinky is a cutting-edge robotics control software, uniquely designed to communicate with hardware through serial communication. It not only responds to user commands, interacts intelligently, and recognizes emotions but also assists in learning robotics, supports studies, and enhances the overall user experience. With autonomous movement and versatile functionalities, Blinky is the ultimate companion for exploration, education, and innovation in robotics.

- **BIONIC PYTHON**

A Python-based 3D hand simulator with realistic finger movements, Arduino sensor integration, and real-time pressure feedback. Features individual finger control, tactile visualization, and comprehensive sensor support.

- **MCP SERVER FOR BROWSER**

This project lets you control a browser (open, click, type, scrape) via a Flask server using Playwright, and uses Google Gemini (Generative AI) to intelligently select and scrape the most relevant link from search results. Remote browser control: Open URLs, click, type, and scrape content via HTTP endpoints. Smart link selection: Uses Gemini AI to pick the best link from search results. Automated scraping: Scrapes and prints the content of the selected page.

- **ASIAN CHICKEN**

Asian Chicken is a 2D endless runner game built with Flutter and the Flame game engine. Guide your chicken (and other characters) through a side-scrolling world, avoid obstacles, and compete for the highest score.

- **COLOR SORTING CAR**

Developed an autonomous color sorting robot with line-following, obstacle avoidance, and object sorting capabilities. Integrated IR sensors for path tracking, ultrasonic sensors for obstacle detection, and a color sensor with servo-actuated mechanisms to identify and sort objects by color.

- **BRACU CORE**

Developed BRACU Core, the mobile version of BRAC University's student portal. Enabled students to access course information, academic records, announcements, and campus resources on-the-go. Focused on user-friendly design, real-time updates, and seamless mobile accessibility to enhance the student experience.

- **PYTHON VIRTUAL ASSISTANT**

This is a simple Python-based virtual assistant that can perform various tasks such as telling the time, creating files, opening applications, capturing selfies, searching the web, and retrieving information from Wikipedia.

- **IMAGE DENOISER**

This is one of my experiment to see how I can manipulate images using opencv and python. This code shows how to denoise image in several methods. Here I've used gaussian blur, median blur, bilateral filter, non local denoise and wavelet denoise method to denoise the image.

- **FIRE FIGHTING ROVER**

Developed an autonomous fire-fighting rover capable of detecting and extinguishing fires. Engineered its mechanical design, control systems, and actuation mechanisms for mobility and firefighting operations. Focused on reliability, real-time responsiveness, and practical deployment in emergency scenarios.

- **HIRE ME APP**

HireMe is a mobile application that bridges the gap between underprivileged workers and the IT world, empowering them with more job opportunities. It also simplifies hiring for users by offering a wide range of services in one app.

- **JOL TORONGO**

Jol Torongo is a underwater robot designed for underwater exploration and mapping. It features a sonar system, depth sensor, and a camera for underwater exploration.

- **GITHUB README GIF GENERATOR**

This project can create gif for your github readme pages with type writer effect. You only have to enter text which you want to have typewriter effect and press capture. It will capture and render a gif with your text so that you can use it in your project.

- **YOUR AI FRIEND**

Your AI Friend is a chat application built using Flutter. It features a neumorphism design style and integrates with the Gemini API for AI-powered chat interactions.

- **FLUTTER MACHINE VISION**

Advanced computer vision application for robotics applications. Implements object detection, tracking, and environment mapping.

- **FOOD BOT**

Food Bot is a Flutter-based Bluetooth app that connects to a Humanoid food serving robot for table and stop position instructions. Data from the list can be saved, and it will automatically go into action. The table list can be added or deleted per user requirement from the UI.

- **SMART HOME AUTOMATION**

Developed a smart home automation system enabling remote control and monitoring of home appliances. Integrated sensors, microcontrollers, and mobile app interfaces to manage lighting, temperature, security, and energy usage. Focused on usability, real-time control, and enhancing convenience and efficiency in residential settings.

- **FLAPPY BIRD**

This is one of my experiment to see how I can manipulate images using opencv and python. This code shows how to denoise image in several methods. Here I've used gaussian blur, median blur, bilateral filter, non local denoise and wavelet denoise method to denoise the image.

- **LEARNERS APP**

learners is a LLM app for our 370 database management system project. learners is a LMS or learning management system app designed for students to make their online learning easy with its minimalistic design and easy to use features.

- **BSTI-TO-UNICODE**

A Python tool to convert Bengali text from BSTI (Bijoy) encoding to Unicode, with optional AI-powered polishing using Google Gemini. Developed by Ratul Hasan (Onnesok).

- **ASSIST ALLY**

Developed Assist Ally, a modern, multi-page Flutter app featuring animated UI, glassmorphic effects, and responsive navigation for mobile and desktop. Implemented full authentication flows, typewriter text effects, multi-phase navigation (Get, Set, Go), and interactive content pages including About Us, Contacts, and Partners. Designed reusable widgets, tabbed navigation, and dynamic backgrounds (images and video) to enhance user experience. Focused on modular project structure, responsiveness, and maintainability.

- **IMAGE STICHER**

Developed an image stitching software to merge multiple images into a seamless panorama. Implemented algorithms for feature detection, alignment, blending, and perspective correction to ensure high-quality results. Focused on accuracy, efficiency, and user-friendly interface for image processing applications.

- **FTP SERVER**

Developed a FTP server using Python to enable secure file transfer and remote storage management. Implemented user authentication, directory browsing, file upload/download, and concurrent client handling. Focused on reliability, security, and efficient network communication.

- **EXPRESSIVE FACE**

Developed Expressive Face App in Flutter to display and animate facial expressions based on user input or predefined triggers. Implemented responsive UI and smooth animations to create an interactive and engaging user experience. Focused on modular design, usability, and cross-platform compatibility.

ACHIEVEMENTS

- **VC AWARD FROM BRAC UNIVERSITY**

Received the Vice Chancellor's Award for outstanding innovation and technical excellence at BRAC University. Recognized for developing advanced projects demonstrating creativity, problem-solving, and practical application of technology. Gained experience in project presentation, mentorship, and implementing solutions that meet high academic and technical standards.

- **IEEE YESIST12 INTERNATIONAL COMPETITOR AND NATIONAL WINNER.**

Represented Onnesok in IEEE YESIST12, competing at the international level and securing the national winner position. Demonstrated innovation, technical expertise, and problem-solving skills in developing bionic arm. Gained experience in global competition standards, presentation, and collaborative teamwork.

- **TECHNOXIAN WORLD ROBOTICS CHALLENGE NATIONAL CHAMPION.**

Won the national championship at the Technoxian World Robotics Challenge, demonstrating advanced skills in robotics design, automation, and problem-solving. Gained experience in high-level competition, teamwork, and innovation under international standards.

- **NATIONAL WINNER OF FIRST BANGLADESHI ONLINE SCIENCE FAIR FROM 10 MINUTE SCHOOL**

Secured the national winner title in Bangladesh's first online science fair organized by 10 Minute School. Demonstrated creativity, scientific understanding, and innovative problem-solving through a Farmbot project.

- **2ND RUNNER-UP IN BDAPPS INNOVATION SUMMIT**

Secured 2nd runner-up position at the BDApps Innovation Summit, showcasing innovative application development and problem-solving skills. Demonstrated creativity, technical expertise, and the ability to develop scalable solutions for real-world challenges.

- **2ND RUNNER-UP AT IOTRIX HACKATHON**

Secured 2nd runner-up position in the IoTrix Hackathon, demonstrating innovation and technical expertise in IoT and smart systems development. Developed a practical solution under tight deadlines, showcasing problem-solving, teamwork, and real-world application of technology. Gained hands-on experience in hackathon-level prototyping, presentation, and collaborative development.

- **NATIONAL WINNER AT DESMOS ART CHALLENGE.**

Awarded national winner at the Desmos Art Challenge, demonstrating creativity, mathematical understanding, and innovative design using the Desmos platform.

- **VERIFIED MICROSOFT AZURE ADMINISTRATOR FROM BANGLADESH HIGH TECH PARK AUTHORITY.**

Verified Microsoft Azure Administrator credential, recognized by the Bangladesh High Tech Park Authority. Demonstrated proficiency in cloud services, infrastructure management, and Azure platform administration.

- **2ND RUNNER UP ON 5TH RUSC SCIENCE FIESTA AND MANY MORE.**

Secured 2nd runner-up position at the 5th RUSC Science Fiesta, showcasing innovation, scientific problem-solving, and project development skills. Also recognized in multiple other competitions for contributions in science and technology.

- **CHAMPION OF 39TH SCIENCE OLYMPIAD**

Crowned champion of the 39th Science Olympiad, demonstrating advanced scientific knowledge, problem-solving skills, and excellence in competitive STEM challenges.

- **WINNER AT UIHP**

Won the UIHP competition with a Bionic Arm project, demonstrating innovation, technical expertise, and practical application of robotics and prosthetic design.

- **INTERNATIONAL GOLD AWARD FOR SENSORY FEEDBACK WITH BIONIC ARM**

Received the International Gold Award for developing a bionic arm with sensory feedback, showcasing innovation, advanced robotics, and practical prosthetic solutions with real-time responsiveness.

- **TECH SPECTRA 2.0 – WINNER**

Secured winner position at Tech Spectra, demonstrating innovation, technical expertise, and problem-solving skills in Robotics.

- **DOUBLE WINNER AT MEDICO ART CHALLENGE**

Won the Medico Art Challenge twice, demonstrating consistent creativity, innovation, and excellence in medical-themed art and design projects.

- **NHSPC REGIONAL WINNER**

Achieved regional winner status in the NHSPC Competitive Programming Contest (CP), demonstrating strong problem-solving, algorithmic thinking, and programming skills.

MORE INFO



<https://www.linkedin.com/in/ratul-hasan-45911b245/>



<https://github.com/Onnesok>



<https://www.facebook.com/ratul.hasan.dev>



<https://youtube.com/Onnesok>



<https://onnesok.github.io/>

PROJECTS LINK

